

Mini Project I — Tourism Analytics

Setup guide: SQL Server → Power BI → Tableau

Work through this once, in order. Budget about 45 minutes, most of it the SQL Server download.

Step 1 — Install SQL Server (about 20 minutes)

You need two free pieces of software.

1a. SQL Server Express (2025 at time of writing)

Download from <https://www.microsoft.com/en-in/sql-server/sql-server-downloads> — scroll to **Express** — Download now.

Run the installer and choose **Basic**. Accept the defaults. When it finishes, the installer shows a **Connection String** box — the instance name is in there. It will almost certainly be:

localhost\SQLEXPRESS

Write it down. Everything downstream uses it.

1b. SQL Server Management Studio (SSMS)

The Express installer has an **Install SSMS** button at the end — use it, or get it from <https://aka.ms/ssmsfullsetup>. Install with defaults.

Check it works: open SSMS. In the connect dialog:

Field	Value
Server type	Database Engine
Server name	localhost\SQLEXPRESS
Authentication	Windows Authentication
Encrypt	Mandatory (the default)
Trust server certificate	tick this

You must tick **Trust server certificate**. A local Express instance uses a self-signed certificate, and with Encrypt set to Mandatory the connection is refused without it. Click **Connect**. If Object Explorer appears on the left, you are done here.

If the server name doesn't work, open **SQL Server Configuration Manager** → SQL Server Services, and read the instance name shown next to "SQL Server".

Step 2 — Build the database (about 5 minutes)

In SSMS, open and execute the five scripts from the sql folder **in order**. Open each with File → Open → File, then press **F5**.

#	Script	What it does	Time
1	01_create_database.sql	Creates TourismDW, 6 dimension tables, a staging table and the fact table	~5 s
2	02_load_data.sql	Loads all 16,865 rows. No CSV files needed — the data is inside the script	30–60 s
3	03_clean_transform_model.sql	Profiles the raw data, cleans it, applies the foreign keys and indexes, creates 4 analytical views	~10 s
4	04_verification_and_analysis.sql	Verification counts and the analytical queries behind the dashboard insights	~5 s
5	05_export_extract_for_tableau.sql	Exports the integrated view as the Tableau extract (see Step 4)	~5 s

Script 2 is 2.4 MB. If SSMS feels sluggish opening it, run it from the command prompt instead — this is faster and uses no memory in SSMS:

```
sqlcmd -S localhost\SQLEXPRESS -E -d master -i "C:\full\path\to\sql\02_load_data.sql"
```

Screenshots to take here

These go straight into the report.

1. **Script 3, Step 1 output** — the raw data quality profile. Shows 60 duplicate rows, 25 negative durations, and NO SHOW sitting in capitals beside the title-case statuses. (The count of distinct statuses reads 4, not 8: SQL Server's default collation is case-insensitive, so it groups COMPLETED with Completed. The casing defect is real and is still standardised during cleaning.)
2. **Script 3, Step 2 output** — raw 12,060 → clean 12,000, 60 rows removed.
3. **Script 4, Section A** — the row counts per table, and the integrity checks all returning 0.
4. **Object Explorer** expanded to show TourismDW — Tables and Views.

What you should see

Table	Rows
DimDate	2,191
DimDestination	32
DimTraveller	2,400
DimAccommodation	155
DimTransport	18
DimBookingChannel	9
stg.FactTrip_Raw	12,060
dbo.FactTrip (clean)	12,000

Step 3 — Open the Power BI project (about 5 minutes)

1. Open **Power BI Desktop**.
2. File → Open → Browse → select `powerbi\TourismAnalytics.pbip`.
3. Power BI will ask for credentials for `localhost\SQLEXPRESS`. Choose **Windows** → *Use my current credentials* → **Connect**.
4. You will then get an **Encryption Support** prompt saying it could not connect using an encrypted connection. Click **OK** to continue unencrypted — same self-signed-certificate situation as SSMS, and fine for a local instance.
5. If Power BI offers to *upgrade the report format to PBIR* or *upgrade the semantic model to TMDL*, choose **Don't upgrade** both times. Those conversions are one-way and the project does not need them.
6. It loads 6 dimension tables plus the fact table. Wait for the refresh to finish — the fact table is 12,000 rows and takes a few seconds.

If your instance is not localhost\SQLEXPRESS: Home → Transform data → Manage Parameters → set **ServerName** to your instance → Close & Apply. That's the only change needed; every query uses the parameter.

7. **Save As** → **TourismAnalytics.pbix**. This is the file you submit.

What's in it

Pages	6
Visuals	50
DAX measures	51, in 9 display folders
Calculated columns	14
Relationships	9 (3 inactive: two alternate date roles used by USERRELATIONSHIP, plus the snowflake link)
Hierarchies	4
Power Query steps	Filtering, type changes, 3 added custom columns on FactTrip

Two things to finish by hand

Drill-through (30 seconds). Go to page **6. Destination Detail**. In the Visualizations pane, find the **Drill through** well and drag `DimDestination` → `DestinationName` into it. Now right-clicking any destination on page 2 offers *Drill through* → *6. Destination Detail*.

Conditional formatting (1 minute). On page 2, select the **Destination scorecard** matrix — Format — Cell elements — turn on **Background colour** for `Avg Satisfaction`, and **Data bars** for `Total Revenue`. On page 3, do the same for the heat map matrix. This is what the rubric means by “appropriate formatting”.

Screenshots to take

One per page, full window, with the filter pane visible. Plus one of the **Model view** showing the star schema, and one of **Power Query Editor** showing the applied steps on `FactTrip`.

Step 4 — Open the Tableau workbook (about 10 minutes)

Read this first if you are on Tableau Public. Tableau Public cannot connect to SQL Server, or to any database. Its connectors are limited to Excel, text/CSV, spatial, statistical and a few web sources — SQL Server, Oracle, MySQL and PostgreSQL are Tableau Desktop only. So on Public you use the **packaged** workbook, which consumes an extract exported from the database rather than connecting to it live.

You have two files.

File	Edition	How it gets the data
<code>TourismAnalytics_Packaged.twbx</code>	Tableau Public or Desktop	Extract exported from <code>dbo.vw_TripAnalysis</code>
<code>TourismAnalytics_SQLServer.twb</code>	Tableau Desktop only	Live Custom SQL on <code>dbo.vw_TripAnalysis</code>

Both contain the same 19 worksheets, 3 dashboards, 18 calculated fields and 4 parameters.

4a. Make the extract genuinely a SQL Server output

Do this so the extract is demonstrably the database’s output rather than a file that merely sits beside it — it is what keeps the “data source” part of the rubric defensible on Public.

1. In SSMS, open and run `sql\05_export_extract_for_tableau.sql`.
2. Tools — Options — Query Results — SQL Server — Results to Grid — tick **Include column headers when copying or saving the results**. Reopen the query window for this to take effect.
3. Run the query, right-click the results grid — **Save Results As...** — `TourismTripAnalysis.csv`.
4. You should get **12,000 rows × 54 columns**, matching the packaged extract exactly. Screenshot the row count — that is Fig. 8.13 in the report.

4b. Open the workbook

1. Open **Tableau Public** → File → Open → `TourismAnalytics_Packaged.twbx`.
2. Screenshot the **Data Source** tab showing the 54 fields.

On Tableau Desktop instead, open `TourismAnalytics_SQLServer.twb`, enter server `localhost\SQLEXPRESS`, database `TourismDW`, **Windows Authentication** → Sign In, and screenshot the Data Source tab with the Custom SQL Query box visible.

Saving on Tableau Public: older versions could only save to the public web. Check **File** → **Save As** on yours — if a local `.twbx` option is offered, use it. If not, save to Tableau Public, set the visualisation to **hidden** in your profile, and download the `.twbx` back from there.

What's in it

Worksheets	19
Dashboards	3
Calculated fields	18, of which 6 are table calculations
Parameters	4
Filters	4 on every worksheet, exposed as controls on the dashboards

Table calculations used: RANK, WINDOW_AVG, RUNNING_SUM / TOTAL, LOOKUP.

Three things to finish by hand

These are quick and they are each worth marks, so don't skip them.

1. Apply the Top N filter (10 seconds). Open the **Top N Destinations** worksheet. From the Data pane, drag **Top N Filter** onto the Filters shelf → tick **True** → OK. The chart now respects the *Top N Destinations* parameter control on dashboard D1.

2. Add two dashboard actions (2 minutes). On **D1 Executive Overview**: Dashboard menu → Actions → Add Action → **Highlight**.

Field	Value
Name	Highlight destination
Source sheets	D1, with <i>Top N Destinations</i> ticked
Target sheets	D1, all
Run action on	Hover
Selected fields	DestinationName

On **D2 Destination & Seasonality**: Actions → Add Action → **Filter**.

Field	Value
Name	Filter detail by destination
Source sheets	D2, <i>Destination Map</i>

Field	Value
Target sheets	D3 Customer & Channel
Run action on	Select
Clearing the selection	Show all values

3. Build the story (3 minutes). Story menu → New Story. Drag dashboard **D1** onto the story point, then click **Blank** to add the next point and drag **D2**, and so on. Type these captions:

Point	Dashboard	Caption
1	D1 Executive Overview	Revenue grew 43% from 2023 to 2026, but 91% of it is international travel
2	D2 Destination & Seasonality	Demand swings 2.2× between the June trough and the December peak
3	D3 Customer & Channel	OTAs take 14–16% commission; direct channels keep 98% of revenue
4	D1 Executive Overview	Loyalty discounts cost margin without buying extra trips per traveller

Rename the story to Tourism Analytics Story.

Then save the workbook. On Tableau Public use File → Save As and take the local .twbx option if it is offered; otherwise save to Tableau Public with the visualisation hidden and download the .twbx back.

Screenshots to take

Data Source tab, each of the 3 dashboards, the story, one worksheet showing the Calculated Field dialog open, and one showing a parameter control in use.

Step 5 — Finish the report

Open Exp8_MiniProject1_Tourism.docx. Every screenshot has a labelled placeholder box telling you exactly what belongs there. Replace the box with your screenshot, keep the caption, and export to PDF.

If something goes wrong

“Cannot connect to localhost”. Open SQL Server Configuration Manager → SQL Server Services → make sure *SQL Server (SQLEXPRESS)* is Running. If it isn’t, right-click → Start.

Power BI: “We couldn’t authenticate with the credentials provided”. File → Options and settings → Data source settings → clear the saved permission for the server, then refresh and pick Windows authentication again.

Power BI: the report opens but visuals show errors. Check the refresh finished. If a single visual complains about a field, it is almost always because the refresh was cancelled partway — Home → Refresh and let it complete.

Tableau: “Unable to connect to the server”. Same check as above, and make sure you typed the instance as localhost\SQLEXPRESS with a backslash.

Tableau: a worksheet is blank. Check the four filter cards on the right — if one has nothing ticked, right-click it — *Select All*.